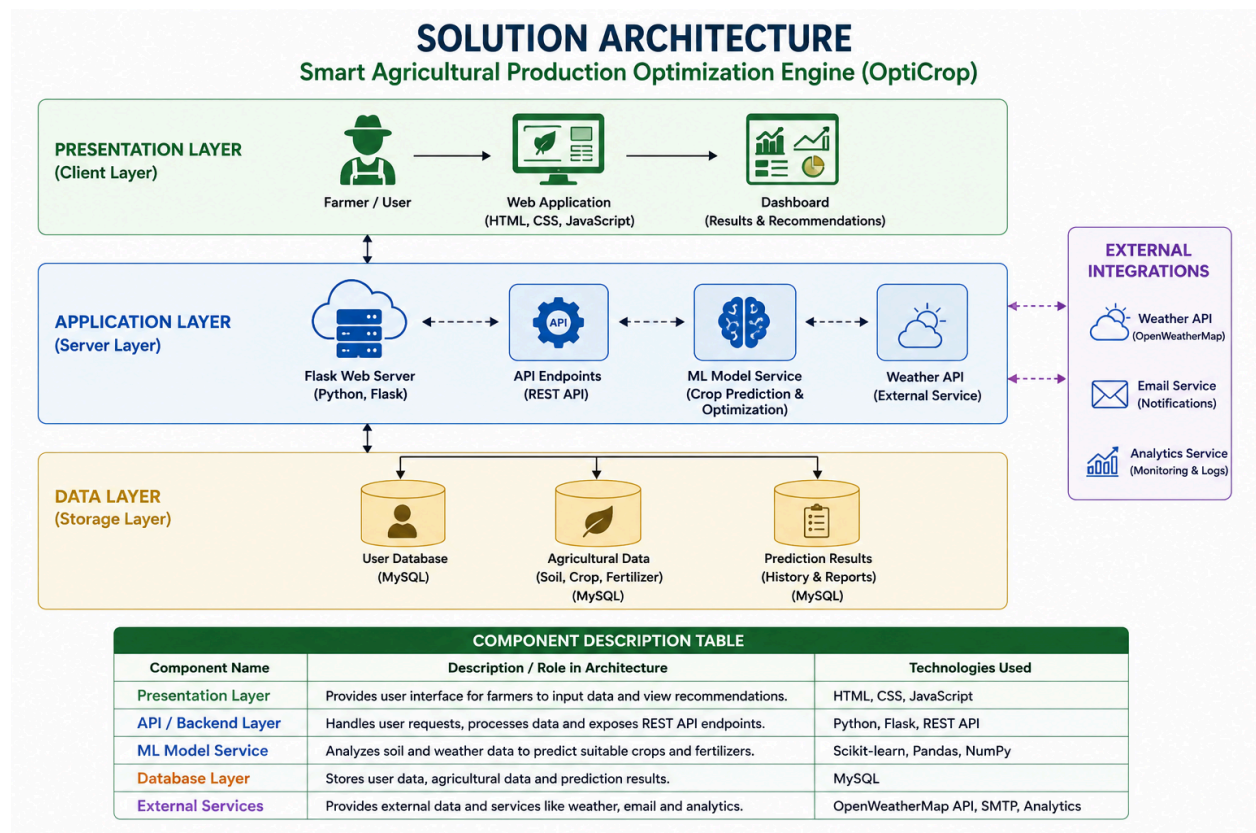


Solution Architecture

Date	8 July 2026
Team ID	
Project name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum marks	5 marks

Solution Architecture Diagram:

The OptiCrop Solution Architecture follows a three-layer architecture consisting of the Presentation Layer, Application Layer, and Data Layer. User inputs are processed by the Flask backend, analyzed using the machine learning model, and stored in the MySQL database. The system then returns crop recommendations, fertilizer suggestions, and production optimization results through the web interface.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User interface for entering agricultural data and viewing recommendations	HTML, CSS, JavaScript
API / Backend	Processes user requests and communicates with the ML model	Python, Flask
Machine Learning Module	Predicts suitable crops and fertilizer recommendations	Scikit-learn, Pandas, NumPy
Database	Stores user information, crop details, and prediction history	MySQL